

Course Code: CSA1208

Course Name: Computer Architecture

Assignment Title	Design and Performance Evaluation of an Intelligent Traffic Management System
Course Outcomes (COs) Covered	CO1: Analyze the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking CO4: Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics. CO5: Construct I/O communication mechanisms by comparing memory-mapped and I/O-mapped techniques, interrupt handling (vectored, hardware, software), DMA controllers, and advanced bus interfaces including PCIe, USB, NVMe, and Thunderbolt.
SDG Mapping	SDG 9 – Industry, Innovation and Infrastructure, SDG 11 – Sustainable Cities and Communities, SDG 13 – Climate Action
BTL	L3 – Apply; L4 – Analyze; L5 – Evaluate

Problem Statement

Design, implement, and evaluate an Intelligent Traffic Management System (ITMS) for a smart, sustainable, and resilient city by integrating real-time data from traffic cameras, vehicle sensors, and road-monitoring devices. The system shall enable inclusive, safe, resilient, and sustainable urban infrastructure by efficiently processing traffic information, optimizing traffic signals, reducing congestion, and supporting priority movement of emergency vehicles.

The proposed system shall analyze suitable processor architecture, instruction sets, instruction formats, and addressing modes for efficient real-time processing of traffic and image-related data. Processor performance shall be evaluated using instruction count, CPI, clock cycles, execution time, and processor utilization, thereby supporting reliable and sustainable infrastructure, technological innovation, and resilient infrastructure development.

The system shall employ an efficient memory organization and cache configuration for high-speed processing of frequently accessed traffic data. Different cache organizations shall be compared using cache hit/miss rate, average memory access time, latency, bandwidth, and throughput to improve energy efficiency and resource utilization. The architecture shall be evaluated under normal traffic, peak-hour congestion, and emergency-vehicle scenarios, considering performance, energy consumption, cost, reliability, scalability, and resilience.

The proposed solution shall be implemented or simulated using suitable tools, with results presented through calculations, tables, graphs, and simulation outputs. The final analysis shall recommend an optimized computer architecture that promotes technological innovation, sustainable and resilient cities, and climate-conscious, energy-efficient infrastructure by reducing traffic congestion, unnecessary energy consumption, and associated environmental impact.

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DESIGN AND PERFORMANCE EVALUATION OF AN INTELLIGENT TRAFFIC MANAGEMENT SYSTEM

INTRODUCTION

The Intelligent Traffic Management System (ITMS) is designed to monitor real-time traffic conditions and optimize traffic signals based on vehicle density. The system uses traffic sensors and cameras as input and processes the collected data using a high-performance processor, hierarchical cache memory, and efficient I/O communication. The system also provides priority to emergency vehicles to improve traffic flow and reduce delays.

OBJECTIVES

The main objectives of the project are:

- To design an intelligent real-time traffic management system.
- To analyze processor performance using instruction count, CPI, clock cycles, and execution time.
- To evaluate L1, L2, L3 cache and main memory performance.
- To optimize traffic signals based on traffic density.
- To provide priority for emergency vehicles.
- To analyze I/O communication using USB, PCIe, DMA, and NVMe.
- To evaluate performance, power, cost, reliability, and scalability.

SYSTEM ARCHITECTURE / DESIGN

➤ Proposed Architecture

The proposed ITMS consists of traffic cameras, vehicle sensors, an I/O unit, processor, cache hierarchy, main memory, traffic analysis software, and traffic signal controllers.

➤ Architecture Flow

Traffic Cameras

+

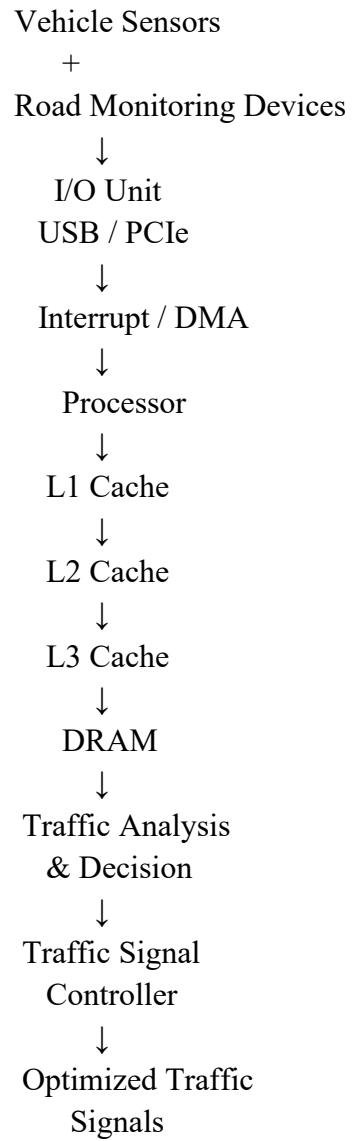
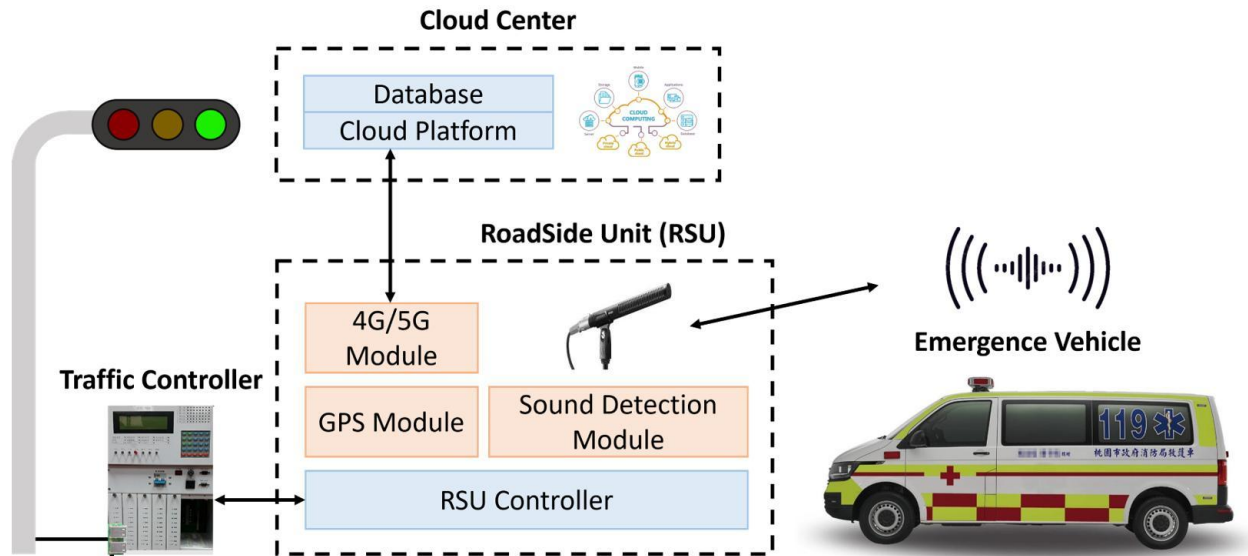


Figure 1: Proposed ITMS Architecture



➤ Working Principle

Traffic cameras and sensors collect vehicle information from different lanes. The input data is transferred through the I/O system to the processor. The processor analyzes the traffic density and calculates suitable green signal timings. Frequently accessed data is stored in the cache hierarchy to reduce memory access time. In case of an emergency vehicle, priority is given to the required lane.

➤ Operating Scenarios

The system is evaluated under three conditions:

1. Normal traffic
2. Peak-hour traffic
3. Emergency vehicle condition

IMPLEMENTATION

The ITMS was implemented using Python. The program performs traffic signal optimization, processor performance calculation, cache performance analysis, and I/O technology evaluation. The implementation uses traffic data for four lanes and calculates signal timing according to the number of vehicles present in each lane.

TRAFFIC WORKLOAD ANALYSIS

➤ Normal Traffic

For normal traffic, the vehicle counts used in the simulation are:

Lane Vehicles

Lane 1 20

Lane 2 15

Lane 3 25

Lane 4 18

The system allocates green signal time according to traffic density.

NORMAL TRAFFIC		
Lane	Vehicles	Green Time
Lane 1	20	35 seconds
Lane 2	15	31 seconds
Lane 3	25	39 seconds
Lane 4	18	33 seconds

Normal Traffic Simulation Output

➤ Peak-Hour Traffic

For peak-hour conditions, higher traffic density is considered:

Lane Vehicles

Lane 1 80

Lane 2 65

Lane 3 95

Lane 4 75

The system provides longer green time to lanes with higher traffic density.

PEAK-HOUR CONGESTION		
Lane	Vehicles	Green Time
Lane 1	80	35 seconds
Lane 2	65	32 seconds
Lane 3	95	38 seconds
Lane 4	75	34 seconds

Peak-Hour Traffic Simulation Output

➤ Emergency Vehicle

An emergency vehicle is detected in Lane 3. The system gives Lane 3 priority by assigning a longer green signal duration.

- Emergency Lane: Lane 3
- Priority Green Time: 60 seconds

This helps reduce the waiting time of emergency vehicles.

```
-----  
EMERGENCY VEHICLE SCENARIO  
-----  
  
Lane 1 : Vehicles = 40 | Green Time = 15 seconds  
Lane 2 : Vehicles = 30 | Green Time = 15 seconds  
Lane 3 : Vehicles = 25 | Green Time = 60 seconds  
Lane 4 : Vehicles = 20 | Green Time = 15 seconds  
  
Emergency vehicle detected in Lane 3.  
Priority green signal activated for Lane 3.  
Lane 3 receives 60 seconds of green time.  
-----
```

Figure

Emergency Vehicle Priority Output

PROCESSOR PERFORMANCE ANALYSIS

➤ Proposed Processor

The proposed processor is evaluated using:

- Instruction Count = 1,000,000
- CPI = 2
- Clock Rate = 2.5 GHz

Calculations

Clock Cycles

$$\begin{aligned}\text{Clock Cycles} &= \text{Instruction Count} \times \text{CPI} \\ &= 1,000,000 \times 2 \\ &= \mathbf{2,000,000 \text{ cycles}}\end{aligned}$$

Execution Time

$$\begin{aligned}\text{Execution Time} &= \text{Clock Cycles} / \text{Clock Rate} \\ &= 2,000,000 / 2.5 \times 10^9 \\ &= \mathbf{0.0008 \text{ seconds}} \\ &= \mathbf{0.8 \text{ ms}}\end{aligned}$$

The processor utilization is considered as **75%**.

➤ **Processor Comparison**

Parameter	Traditional	Proposed
Instruction Count	1,500,000	1,000,000
CPI	3	2
Clock Rate	2.0 GHz	2.5 GHz
Clock Cycles	4,500,000	2,000,000
Execution Time	2.25 ms	0.8 ms
Utilization	85%	75%

The proposed processor requires fewer clock cycles and provides lower execution time than the traditional architecture.

PROPOSED PROCESSOR

```
-----  
Instruction Count      : 1000000  
CPI                   : 2  
Clock Rate            : 2.5 GHz  
Clock Cycles          : 2000000  
Execution Time        : 0.0008 seconds  
Execution Time        : 0.8 ms  
Processor Utilization : 75 %
```

CONVENTIONAL PROCESSOR

```
-----  
Instruction Count      : 1500000  
CPI                   : 3  
Clock Rate            : 2.0 GHz  
Clock Cycles          : 4500000  
Execution Time        : 2.25 ms  
Processor Utilization : 85 %
```

Processor Execution-Time Improvement: 64.44 %

Processor Performance Output

MEMORY AND CACHE PERFORMANCE

The proposed system uses a hierarchical cache consisting of L1, L2, and L3 caches followed by DRAM.

The simulation considers 1,000 cache accesses.

Parameter	Value
Cache Accesses	1000
Cache Hits	920
Cache Misses	80
Hit Rate	92%
Miss Rate	8%
L1 Latency	1 ns
L2 Latency	4 ns
L3 Latency	12 ns
DRAM Latency	100 ns

Parameter	Value
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Simplified AMAT	1.32 ns
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Cache Hit Rate

Hit Rate = (Cache Hits / Total Accesses) × 100

= (920 / 1000) × 100

= **92%**

Cache Miss Rate

Miss Rate = (80 / 1000) × 100

= **8%**

Simplified AMAT

AMAT = L1 Latency + Miss Rate × L2 Latency

= 1 + (0.08 × 4)

= **1.32 ns**

The high cache hit rate reduces the number of accesses to slower DRAM and improves overall processor performance.

CACHE PERFORMANCE

Total Cache Accesses	: 1000
Cache Hits	: 920
Cache Misses	: 80
Cache Hit Rate	: 92.0 %
Cache Miss Rate	: 8.0 %

MEMORY HIERARCHY

L1 Cache Latency	: 1 ns
L2 Cache Latency	: 4 ns
L3 Cache Latency	: 12 ns
DRAM Latency	: 100 ns

Average Memory Access Time (AMAT): 1.32 ns

Cache Performance Output

MEMORY TECHNOLOGY COMPARISON

Memory	Performance	Main Application
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SRAM	Very High	L1/L2/L3 Cache
DRAM	High	Main Memory
NAND Flash	Medium	Data Storage
MRAM	High	Persistent Memory
RRAM	High	Emerging Memory Systems

For the proposed ITMS, SRAM is suitable for cache, while DRAM is used as main memory. NAND Flash/NVMe storage can be used for long-term traffic data storage.

I/O COMMUNICATION ANALYSIS

Efficient I/O communication is required to transfer data from traffic cameras and sensors to the processing system.

Technology	Application
USB	Sensor communication
PCIe	High-speed camera/data transfer
DMA	Direct data transfer with reduced CPU involvement
NVMe	High-speed traffic data storage

DMA is useful for transferring large amounts of camera and sensor data while reducing processor workload. Interrupts are used for important events such as emergency vehicle detection.

SYSTEM-LEVEL PERFORMANCE EVALUATION

The proposed system is evaluated based on the following factors:

Factor	Evaluation
Performance	High-speed real-time processing
Memory Efficiency	92% cache hit rate
Power	Reduced CPU involvement using DMA
Cost	Balanced memory and processor design
Reliability	Continuous traffic monitoring
Scalability	Supports additional lanes and sensors
Sustainability	Reduced vehicle waiting time

The system provides better response for real-time traffic conditions while maintaining efficient processor and memory usage.

ENGINEERING RECOMMENDATION

Based on the analysis, the following architecture is recommended:

- 2.5 GHz high-performance processor

- L1, L2, and L3 hierarchical cache
- DRAM main memory
- DMA-based I/O transfer
- PCIe for high-speed communication
- NVMe for traffic data storage
- Dynamic traffic signal optimization
- Emergency vehicle priority mechanism

This combination provides faster processing, efficient memory access, and improved traffic management.

SUPPORTING TOOL / SIMULATION EVIDENCE

The system was implemented and tested using Python.

The simulation generates outputs for:

- Normal traffic
- Peak-hour traffic
- Emergency vehicle priority
- Processor performance
- Cache performance
- I/O communication

SYSTEM-LEVEL EVALUATION	
Performance	: High-speed real-time processing
Memory Efficiency	: 92% cache hit rate
Power Efficiency	: Reduced CPU involvement using DMA
Cost	: Balanced cache and main-memory design
Reliability	: Continuous traffic monitoring
Scalability	: Supports additional lanes and sensors
Sustainability	: Reduced congestion and vehicle waiting time

SDG MAPPING

SDG	Contribution
SDG 9 – Industry, Innovation and Infrastructure	Uses intelligent computing and smart infrastructure.

SDG 11 – Sustainable Cities and Communities	Helps reduce traffic congestion and improve transportation.
SDG 13 – Climate Action	Reduced waiting time can help reduce unnecessary fuel consumption and emissions.

CONCLUSION

The Intelligent Traffic Management System was designed and implemented to demonstrate how computer architecture concepts can be applied to real-time traffic management. The simulation successfully analyzes traffic conditions, optimizes signal timing, and provides priority to emergency vehicles.

The processor analysis achieved **2,000,000 clock cycles** and **0.8 ms execution time**, while the cache analysis achieved a **92% hit rate** and a simplified **AMAT of 1.32 ns**. The use of hierarchical cache, efficient I/O, DMA, and high-speed communication technologies improves the overall responsiveness of the system.

Therefore, the proposed architecture provides an efficient and scalable solution for intelligent traffic management while supporting the goals of **SDG 9, SDG 11, and SDG 13**.

INDIVIDUAL CONTRIBUTION

S. No.	Name	Reg. No.	Contribution
1	D. Gnana Deepthi	192411123	Processor Performance Analysis
2	Shruthi K.	192571019	Memory & Cache Optimization
3	Suryasri S.	192511218	Traffic Workload & System-Level Evaluation